Dimensionality Reduction

- **What Is Dimensionality Reduction?**
- **Dimensionality Reduction Methods**
  - Principal Component Analysis
  - Attribute Subset Selection
  - Nonlinear Dimensionality Reduction Methods

What Is Dimensionality Reduction?

- **Curse of dimensionality**
  - When dimensionality increases, data becomes increasingly sparse
  - Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
  - The possible combinations of subspaces will grow exponentially
- **Dimensionality reduction**
  - Reducing the number of random variables under consideration, via obtaining a set of principal variables
- **Advantages of dimensionality reduction**
  - Avoid the curse of dimensionality
  - Help eliminate irrelevant features and reduce noise
  - Reduce time and space required in data mining
  - Allow easier visualization

Dimensionality Reduction Methods

- **Dimensionality reduction methodologies**
  - **Feature selection**: Find a subset of the original variables (or features, attributes)
  - **Feature extraction**: Transform the data in the high-dimensional space to a space of fewer dimensions
- **Some typical dimensionality reduction methods**
  - Principal Component Analysis
  - Attribute Subset Selection
  - Nonlinear Dimensionality Reduction